

Symplectic Manifolds

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Abstract

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1 The Symplectic Form

1.1 Basic Constructions

Definition 1.1.1 (Symplectic Manifolds)

Let M be a smooth manifold. Let $\omega \in \Omega^2(M)$ be a 2-form. We say that ω is a symplectic form if the following are true.

- ω is a symplectic structure on TM .
- $d\omega = 0$.

In this case we call (M, ω) a symplectic manifold.

Lemma 1.1.2

Let (M, ω) be a symplectic manifold. Then $\dim(M)$ is even.

Definition 1.1.3 (The Liouville Volume Form)

Let (M, ω) be a symplectic manifold of dimension $2n$. Define the Liouville volume form of M by

$$\mu_M = \frac{\omega^n}{n!}$$

Proposition 1.1.4

Let (M, ω) be a symplectic manifold. Then the following are true.

- μ_M is a nowhere-vanishing differential form.
- M is orientable.

Example 1.1.5

The following are symplectic manifolds.

- $\mathbb{R}^{2n}$ with coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$ and with symplectic form

$$\sum_{i=1}^n dx_i \wedge dy_i$$

- For all $g \geq 1$, the compact orientable surface Σ_g with any ω such that $0 \neq [\omega] \in H_{\text{dR}}^2(\Sigma_g)$.

1.2 Standard Symplectic Coordinates

Theorem 1.2.1 (Moser's Theorem Relative Version)

Let M be a smooth manifold. Let X be an embedded compact submanifold of M . Let $\iota : X \hookrightarrow M$ be the inclusion map. Let ω_0 and ω_1 be symplectic forms on M . If $\omega_0|_X = \omega_1|_X$, then there exists $X \subseteq U_0, U_1 \subseteq M$ and a map $\varphi : U_0 \rightarrow U_1$ such that the following are true.

- $\iota = \varphi \circ \iota$.
- φ is a diffeomorphism.
- $(\varphi|_X)^* \omega_1 = \omega_0$.

Theorem 1.2.2 (Darboux (1982))

Let (M, ω) be a symplectic manifold. Let $p \in M$. Then there exists a chart $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ at p such that

$$\omega|_U = \sum_{i=1}^n dx^i \wedge dy^i$$

Definition 1.2.3 (Darboux Coordinates)

Let (M, ω) be a symplectic manifold. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a local chart of M . We say

that the chart is Darboux if

$$\omega|_U = \sum_{i=1}^n dx^i \wedge dy^i$$

1.3 Maps Preserving the Symplectic Structure

Definition 1.3.1 (Symplectic Maps)

Let (M, ω) and (N, σ) be symplectic manifolds. Let $f : M \rightarrow N$ be a smooth map.

- We say that f is symplectic if

$$f^* \sigma = \omega$$

- We say that f is a symplectomorphism if f is symplectic and a diffeomorphism.

Lemma 1.3.2

Let (M, ω) and (N, σ) be symplectic manifolds. Let $f : M \rightarrow N$ be a symplectic map. Then f is an immersion.

Definition 1.3.3 (The Group of Symplectomorphisms)

Let (M, ω) be a symplectic manifold. Define the group of symplectomorphisms of M to be the subgroup

$$\text{Symp}(M, \omega) = \{ f : M \rightarrow M \mid f \text{ is a symplectomorphism} \} \leq \text{Diff}(M)$$

1.4 Symplectic Structure on the Cotangent Bundle

Through out this section we denote the projection map $T^*M \rightarrow M$ by π_M .

Definition 1.4.1 (The Canonical Hamiltonian Structure)

Let M be a smooth manifold. Define the canonical Hamiltonian structure ω on T^*M as follows. At the local chart $(U, q^1, \dots, q^n, p^1, \dots, p^n)$, ω is given by

$$\omega|_U = \sum_{i=1}^n dq^i \wedge dp^i$$

Lemma 1.4.2

Let M be a smooth manifold. Let ω be the canonical Hamiltonian structure. Then (T^*M, ω) is a symplectic manifold.

Definition 1.4.3 (The Liouville 1-Form)

Let M be a smooth manifold. Let $\pi : T^*M \rightarrow M$ be the bundle map. Define the Liouville 1-form $\theta \in \Omega^1(T^*M)$ as follows. For each $q \in M$ and $p \in T_q^*M$, define

$$\theta_{(q,p)} = p(\pi_M(v))$$

for any $v \in T_{(q,p)}(T^*M)$.

Proposition 1.4.4

Let M be a smooth manifold. Let $(U, q^1, \dots, q^n, p^1, \dots, p^n)$ be a chart of T^*M where $(q_1, \dots, q_n) \in M$. Then θ is given in local coordinates by

$$\theta|_U = \sum_{i=1}^n p^i dq^i$$

Proposition 1.4.5

Let M be a smooth manifold. Let θ be the Liouville 1-form. Let ω be the canonical Hamiltonian structure. Then we have

$$\omega = -d\theta$$

Definition 1.4.6 (The Lift of a Diffeomorphism)

Let M, N be smooth manifolds. Let $f : M \rightarrow N$ be a diffeomorphism. Define the lift of f to be the map $f_{\#} : T^*M \rightarrow T^*N$ given by

$$f_{\#}(p, v) = (f(p), w)$$

where $w \in T_p^*N$ is the unique element such that $v = (df_p)^*w$.

Proposition 1.4.7

Let M, N be smooth manifolds. Let $f : M \rightarrow N$ be a diffeomorphism. Then the following are true.

- We have $f \circ \pi_M = \pi_N \circ f_{\#}$.
- $f_{\#}^* \theta_N = \theta_M$.

2 Lagrangian Submanifolds

2.1 Basic Constructions

Definition 2.1.1 (Lagrangian Vector Subspaces)

Let k be a field. Let (V, ω) be a symplectic vector space over k . Let $W \subseteq V$ be a subspace. We say that W is Lagrangian if the following are true.

- $\omega|_W = 0$.
- $2 \dim(W) = \dim(V)$.

Definition 2.1.2 (Lagrangian Submanifolds)

Let (M, ω) be a symplectic manifold. Let L be an embedded submanifold of M . We say that L is Lagrangian if $T_p L$ is a Lagrangian subspace of $T_p M$.

Lemma 2.1.3

Let (M, ω) be a symplectic manifold. Let L be an embedded submanifold of M . Then L is Lagrangian if and only if the following are true.

- $\iota^* \omega = 0$ where $\iota : L \hookrightarrow M$ is the inclusion map.
- $2 \dim(L) = \dim(M)$.

Proposition 2.1.4

Let M be a smooth manifold. Let $s \in \Omega^1(M)$. Then $\text{im}(s)$ is a Lagrangian submanifold of T^*M if and only if $ds = 0$.

Theorem 2.1.5 (Weinstein Lagrangian Neighborhood Theorem)

Let M be a smooth manifold. Let X be an embedded compact submanifold of M . Let $\iota : X \hookrightarrow M$ be the inclusion map. Let ω_0 and ω_1 be symplectic forms on M such that X is a Lagrangian submanifold of (M, ω_0) and (M, ω_1) . Then there exists $X \subseteq U_0, U_1 \subseteq M$ and a map $\varphi : U_0 \rightarrow U_1$ such that the following are true.

- $\iota = \varphi \circ \iota_*$.
- φ is a diffeomorphism.
- $(\varphi|_X)^* \omega_1 = \omega_0$.

2.2 Relation to Symplectomorphisms

Proposition 2.2.1

Let (M, ω) and (N, σ) be symplectic manifolds. Let $f : M \rightarrow N$ be a diffeomorphism. Then f is a symplectomorphism if and only if Γ_f is a Lagrangian submanifold of $(M \times N, \text{proj}_M^* \omega - \text{proj}_N^* \sigma)$.

Recipe for producing symplectomorphisms.

1. Start with a Lagrangian submanifold of $(T^*M \times T^*N, \omega \times \sigma)$.
2. Send the Lagrangian through the twist map $\tau : T^*M \times T^*N \rightarrow T^*M \times T^*N$ given by $(x, v) \mapsto (x, -v)$. The image is then a Lagrangian.
3. Check whether $\tau(\sigma)$ is the graph of some diffeomorphism $\varphi : M \rightarrow N$.
4. If it is, then φ is a symplectomorphism.

2.3 Generating Functions

Definition 2.3.1 (Generating Functions)

Let M, N be smooth manifolds. Let $f \in \mathcal{C}^\infty(M \times N)$. Define

$$Y_f = \{(p, q, \text{proj}_{T_p^* M} d_{(p,q)} f, -\text{proj}_{T_q^* M} d_{(p,q)} f) \in M \times N \times T^*M \times T^*N \mid (p, q) \in M \times N\}$$

We say that f is a generating function if there exists a diffeomorphism $\varphi : T^*M \rightarrow T^*N$ such that

$$Y_f = \Gamma_\varphi.$$

In this case, we know that $\varphi : T^*M \rightarrow T^*N$ is given by

$$\varphi(x, \text{proj}_{T_p^*M} d_{(p,q)} f) = (y, -\text{proj}_{T_q^*M} d_{(p,q)} f)$$

3 Calculus on Symplectic Manifolds

3.1 Symplectic Vector Fields

Definition 3.1.1 (Symplectic Vector Fields)

Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a vector field. We say that X is symplectic if $\iota_X(\omega)$ is a closed form.

Proposition 3.1.2

Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a vector field. Let θ^X be the flow of X . Then X is symplectic if and only if $\theta^X(t, -)$ is a symplectomorphism for all t .

3.2 Hamiltonian Vector Fields

Definition 3.2.1 (The Vector Field of a Function)

Let (M, ω) be a symplectic manifold. Let $H \in \mathcal{C}^\infty(M)$. Define the associated vector field of H to be the unique vector field $X_H \in \mathfrak{X}(M)$ such that

$$\iota_{X_H}(\omega) = df$$

where ι is the interior product.

Proposition 3.2.2

Let (M, ω) be a symplectic manifold. Let $H \in \mathcal{C}^\infty(M)$. Let $(U, x^1, \dots, x^n, y^1, \dots, y^n)$ be a Darboux chart. Then the vector field of H is given locally at the chart by

$$(X_H)|_U = \sum_{i=1}^n \left(\frac{\partial H}{\partial y^i} \frac{\partial}{\partial x^i} - \frac{\partial H}{\partial x^i} \frac{\partial}{\partial y^i} \right)$$

Definition 3.2.3 (Hamiltonian Vector Fields)

Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a vector field. We say that X is Hamiltonian if there exists a smooth map $H \in \mathcal{C}^\infty(M)$ such that

$$X = X_H$$

Proposition 3.2.4

Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a vector field. If X is Hamiltonian, then X is symplectic.

Theorem 3.2.5 (Liouville's Theorem)

Let (M, ω) be a symplectic manifold. Let $X \in \mathfrak{X}(M)$ be a complete Hamiltonian vector field. Let θ^X be the flow of X . Then we have

$$\theta^X(t, -)^* \mu_M = \mu_M$$

3.3 The Poisson Structure

Definition 3.3.1 (The Poisson Bracket)

Let (M, ω) be a symplectic manifold. Define the poisson bracket of M to be the map $\{-, -\} : \mathcal{C}^\infty(M) \times \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M)$ by

$$\{f, g\} = X_f(g)$$

Lemma 3.3.2

Let (M, ω) be a symplectic manifold. Then the following are true.

- $(M, \{-, -\})$ is a Lie algebra.
- Leibniz rule: $\{f, gh\} = \{f, g\}h + g\{f, h\}$.

Proposition 3.3.3

Let (M, ω) be a symplectic manifold. Then the map $\mathcal{C}^\infty(M) \rightarrow \mathfrak{X}(M)$ given by $f \mapsto X_f$ is a Lie algebra homomorphism.

3.4 Integrable Systems**Definition 3.4.1 (Hamiltonian Systems)**

A Hamiltonian system is a triple (M, ω, H) where (M, ω) is a symplectic manifold and $H \in \mathcal{C}^\infty(M)$.

Definition 3.4.2 (Integrable Systems)

Let (M, ω, H) be a Hamiltonian system. Suppose that $\dim(M) = 2n$. We say that it is integrable if there exists $f_i \in \mathcal{C}^\infty(M)$ for $1 \leq i \leq n$ and $f_1 = H$ such that

$$\{f_i, f_j\} = 0$$

for all i, j .

Theorem 3.4.3 (Arnold-Liouville Theorem)

Let $(M, \omega, H, f_2, \dots, f_n)$ be an integrable system. Let $c \in \mathbb{R}$. Then $(H, f_2, \dots, f_n)^{-1}(c)$ is a Lagrangian submanifold of M . Moreover, if M is compact and connected, then the following are true.

- $(H, f_2, \dots, f_n)^{-1}(c)$ is diffeomorphic to T^k for some $k \in \mathbb{N}^+$.
- Angle coordinates.

4 Symplectic and Hamiltonian Actions

4.1 Symplectic Actions

Definition 4.1.1 (Symplectic Actions)

Let (M, ω) be a symplectic manifold. Let G be a Lie group. Let $\varphi : G \rightarrow \text{Diff}(M)$ be a smooth action. We say that the action is symplectic if

$$\text{im}(\varphi) \subseteq \text{Symp}(M)$$

Lemma 4.1.2

Let (M, ω) be a symplectic manifold. Then there is a one to one bijection

$$\{X \in \mathfrak{X}(M) \mid X \text{ is symplectic and complete}\} \xleftrightarrow{1:1} \{\varphi : \mathbb{R} \times M \rightarrow M \mid \varphi \text{ is a symplectic action}\}$$

given as follows.

- For a complete and symplectic vector field X , the map $t \mapsto \exp(tX)$ is a symplectic action.
- For a symplectic action φ , the map $p \mapsto \frac{d\varphi(t,p)}{dt}$ is a complete and symplectic vector field.

4.2 Moment Maps

Definition 4.2.1 (Moment Maps)

Let (M, ω) be a symplectic manifold. Let G be a Lie group with a symplectic action on M . A moment map of M is a map

$$\mu : M \rightarrow \text{Lie}(G)^* \cong \mathbb{R}^{\dim(G)}$$

such that the following are true.

- Hamilton's condition: For all $\xi \in \text{Lie}(G)$, the fundamental vector field ξ_M is Hamiltonian via the map $\langle \mu, \xi \rangle : M \rightarrow \mathbb{R}$.
- Equivariance: For all $g \in G$, we have

$$\mu(g \cdot x) = \text{Ad}_g^*(\mu(x))$$

for all $x \in M$.

Definition 4.2.2 (Hamiltonian Actions)

Let (M, ω) be a symplectic manifold. Let G be a Lie group with a symplectic action on M . We say that the action is Hamiltonian if it admits a moment map $\mu : M \rightarrow \text{Lie}(G)^*$.

4.3 Symplectic Quotients

Definition 4.3.1 (Regular Values)

Let (M, ω) be a symplectic manifold. Let G be a Lie group with a Hamiltonian action on M . Let $\xi \in \text{Lie}(G)^*$. We say that ξ is a regular value if the induced map

$$d\mu_p : T_p M \rightarrow \text{Lie}(G)^*$$

is surjective for all $p \in \mu^{-1}(\xi)$.

Definition 4.3.2 (Coadjoint Stabilizer)

Let (M, ω) be a symplectic manifold. Let G be a Lie group with a Hamiltonian action on M . Let $\xi \in \text{Lie}(G)^*$ be a regular value. Define the coadjoint stabilizer of ξ by

$$G_\xi = \{g \in G \mid \text{Ad}_g^*(\xi) = \xi\}$$

Definition 4.3.3 (Symplectic Quotients)

Let (M, ω) be a symplectic manifold. Let G be a Lie group with a Hamiltonian action on M . Let $\xi \in \text{Lie}(G)^*$ be a regular value. Define the symplectic quotient of M by G to be the orbit space

$$M//G = \frac{\mu^{-1}(\xi)}{G_\xi}$$

Theorem 4.3.4 (The Marsden–Weinstein–Meyer Theorem)

Let (M, ω) be a symplectic manifold. Let G be a Lie group with a Hamiltonian action on M . Let $\xi \in \text{Lie}(G)^*$ be a regular value. If G_ξ acts freely and properly on $\mu^{-1}(\xi)$, then $(M//G, \bar{\omega})$ is a symplectic manifold where $\bar{\omega}$ is given by

$$\iota^* \omega = \pi^* \bar{\omega}$$

where $\iota : \mu^{-1}(\xi) \hookrightarrow M$ is the inclusion map and $\pi : \mu^{-1}(\xi) \rightarrow M//G$ is the quotient map.

Proposition 4.3.5

Let (M, ω) be a symplectic manifold. Let G be a Lie group with a Hamiltonian action on M . Let $\xi \in \text{Lie}(G)^*$ be a regular value. Suppose that G_ξ acts freely and properly on $\mu^{-1}(\xi)$. Then we have

$$\dim(M//G) = \dim(M) - 2 \dim(G_\xi)$$

4.4 Hamiltonian Torus Actions**Theorem 4.4.1** (The Atiyah–Guillemin–Sternberg convexity theorem (1982))

Let (M, ω) be a compact connected symplectic manifold. Let $T^k \times M \rightarrow M$ be a Hamiltonian action. Then $\mu(M)$ is a convex polytope in $\text{Lie}(T^k)^*$ given by

$$\mu(M) = \text{Conv}(\text{Fix}_{T^k}(M))$$

Definition 4.4.2 (Toric Symplectic Manifold)

Let (M, ω) be a symplectic manifold of dimension $2n$. Let $T^n \times M \rightarrow M$ be a Hamiltonian torus action. We say that M is a toric symplectic manifold if M is compact connected and the action is faithful.

Delzant polytope: simple, rational, smooth

Theorem 4.4.3 (Delzant (1988))

There is a one to one bijection

$$\frac{\{(M, \omega) \mid (M, \omega) \text{ is a toric symplectic manifold}\}}{\cong} \xleftrightarrow{1:1} \{P \mid P \text{ is a Delzant polytope}\}$$